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ENGINEERING:E043

MCA:C570

MBA: B181

M.TECH: T834

PARKINSON'S DISEASE DETECTION USING MACHINE LEARNING TECHNIQUES

CSE(DATA SCIENCE)

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Introduction

- Parkinson's disease is a progressive neurodegenerative disorder that mainly affects the motor functions of the human body.
 - It causes symptoms such as tremors, muscle stiffness, slow movement, and postural instability.
- The disease occurs due to the loss of dopamine-producing neurons in the brain.
- Patients may also experience non-motor symptoms like depression, cognitive decline, and speech problems.
- Early detection is difficult because symptoms develop gradually over time.
- Machine learning helps analyze biomedical data and enables early prediction of Parkinson's disease.

Problem Statement

- Traditional diagnosis of Parkinson's disease depends on clinical observation and neurological tests.
- These methods are time-consuming and may not detect the disease in early stages.
 - Early symptoms are mild and often difficult to identify accurately.
 - Manual analysis of medical data is complex and inefficient.
- There is a need for an automated system that can provide fast and accurate diagnosis.

Objectives

The main objective is to develop a machine learning model to detect Parkinson's disease.

The system uses voice-based biomedical features for prediction.

It aims to improve accuracy and reduce diagnosis time.

The model is designed to be reliable and efficient.

It also helps doctors in making better clinical decisions.

Methodology

The dataset is collected from the UCI Machine Learning Repository.

It contains 195 samples and 23 voice-related features.

Data preprocessing is performed to clean and prepare the dataset.

Feature scaling is applied to normalize the values.

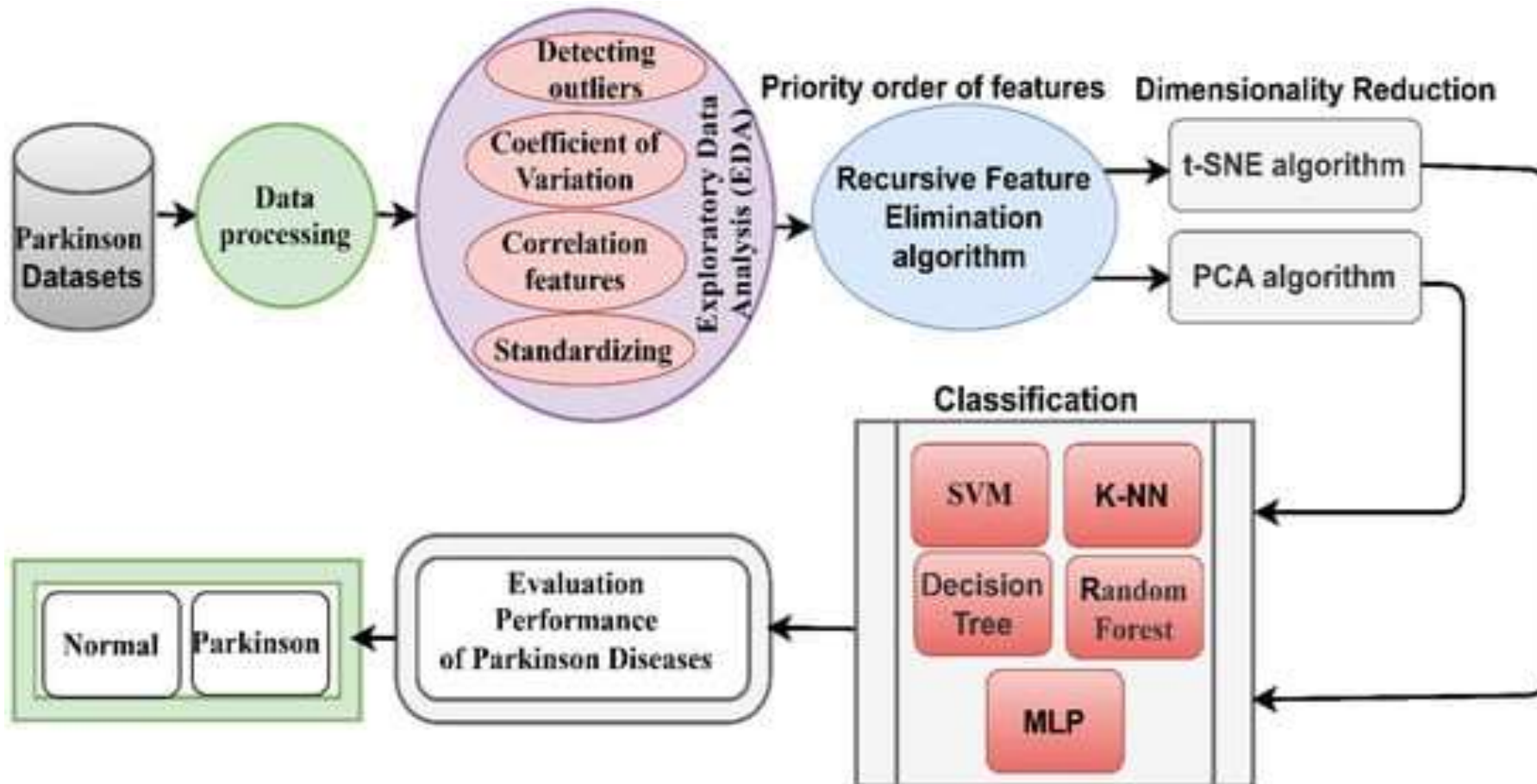
Important features are selected to improve model performance.

The dataset is divided into training and testing sets.

Random Forest and XGBoost algorithms are used for classification.

Model performance is evaluated using accuracy, precision, recall, and F1-score.

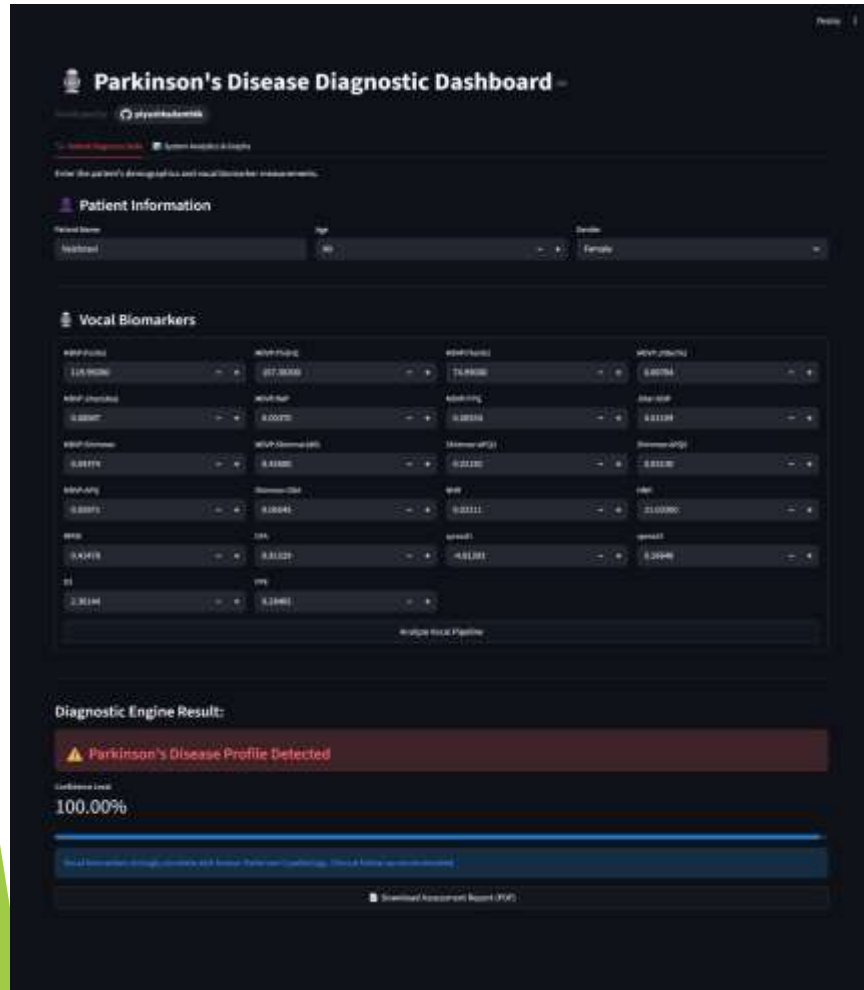
System Architecture



Results

- ▶ The proposed system shows high accuracy in detecting Parkinson's disease.
 - ▶ Random Forest achieves around 92% accuracy.
 - ▶ XGBoost provides better performance with about 94% accuracy.
- ▶ A confusion matrix is used to analyze model performance.
 - ▶ The system shows good precision and recall values.
- ▶ It can accurately classify healthy and affected individuals.

Results



Conclusion

- The project shows that machine learning can detect Parkinson's disease effectively.
- Voice features are useful indicators for prediction.
- The system achieves high accuracy and reliability.
 - It can assist doctors in early diagnosis.
- Overall, it improves the efficiency of medical diagnosis.

Future Work

- ▶ Deep learning models can be used to improve performance.
- ▶ A real-time web or mobile application can be developed.
 - ▶ More datasets can be added for better accuracy.
 - ▶ Other data types like gait and handwriting can be included.
- ▶ The system can be tested in real-world healthcare environments

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Thank You